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August 10, 2026

The Editors
Journal of Cheminformatics
BioMed Central

Dear Editors,

We are pleased to submit our manuscript, “**Do larger molecular models improve antimalarial activity prediction? A scaffold-controlled comparison with ECFP4**”, for consideration as a Research Article in the *Journal of Cheminformatics*.

The manuscript asks whether graph neural networks and a pretrained sequence transformer improve activity ranking beyond a compact ECFP4-random-forest baseline when chemical scaffolds are held out. We test this question on a frozen panel of 19 836 African antimalarial natural products under random and Bemis–Murcko scaffold cross-validation (25 fold–seed replicates), comparing GIN, GIN–TFP, GIN–TNE, and ChemBERTa with ECFP4–RF.

Under the scaffold split, the practically relevant scaffold-held-out setting, **ECFP4–RF achieves the highest ROC AUC (0.830 0)**, ahead of GIN–TFP (0.813 8), GIN–TNE (0.809 0), GIN (0.804 7) and ChemBERTa (0.786 7); all four learned arms are significantly lower after multiplicity correction. Under the random split the ordering is unchanged: ECFP4–RF reaches 0.943 3, versus 0.912 1 for ChemBERTa, 0.909 8 for GIN, 0.908 4 for GIN–TFP and 0.891 8 for GIN–TNE (all paired $p < 0.000 1$ after Benjamini–Hochberg correction). The comparison shows that the fingerprint baseline remains strongest under scaffold-held-out testing. The topological fusion arm narrows the gap modestly, while projection-weight analysis identifies persistent-image dimensions as a salient block; this is presented as descriptive evidence about representation use, not causal attribution. The fold-independent initialization protocol is specified so that transformer performance is not inflated by cross-fold weight transfer.

As an external transfer analysis, we repeated the ECFP4–RF versus GIN comparison on a molecule-disjoint ChEMBL-derived *P. falciparum* activity panel ($n = 22\,267$ after exclusion of 180 canonical-SMILES overlaps). ECFP4–RF again outperformed GIN under the random (0.954 7 versus 0.923 7) and scaffold (0.919 0 versus 0.884 3) splits. This supports, but does not universalise, the conclusion beyond the internal screening labels; the ChEMBL labels are thresholded IC₅₀/EC₅₀ records rather than prospective experiments. The study therefore offers a controlled benchmark and a reproducible negative result within the scope of the *Journal of Cheminformatics*. Code, frozen splits and trained checkpoints are available in the public repository (https://github.com/NanaEngo/Malaria_codesV2).

This work is original, is not under consideration elsewhere, and all authors have approved the

submission.

We thank you for your consideration.

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Suggested reviewers:

- Prof. Gisbert Schneider (generative molecular design, fingerprints)
- Dr. Ola Engkvist (de novo design, graph models)
- Dr. Davide Boldini (natural-product fingerprints and chemical space)